

UCS745: CLOUD & DEVOPS

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Course Objective: The objective of the course is to teach techniques to automate the process of integration and deployment software product.

Syllabus

Introduction to DevOps: Definition of DevOps, Challenges of traditional IT systems & processes, History and emergence of DevOps, DevOps definition and principles governing DevOps, DevOps and Agile, The need for building a business use case for DevOps, Purpose of DevOps, Application Deployment, Automated Application Deployment, Application Release Automation (ARA), Components of Application Release Automation (ARA), Continuous Integration, Best Practices of CI, Benefits of CI, Continuous Delivery

Typical Toolkit for DevOps: DevOps, An Overview, Achieving DevOps, Continuous Practices, Continuous Integration (CI), How does CI Work? Continuous Integration Practices, Benefits of Continuous Integration A Quick Recap of Continuous Delivery, Continuous Delivery Process, Benefits of Continuous Delivery, Continuous Deployment

Source Code Management: History of Version Control Systems (VCS), Basic operations in a VCS, Examples of version control systems, Subversion (SVN), Features and Limitations, Mercurial, Git, Overview, History - Linux and Git by Linus Torvalds, Advantages of Git, Explain how local version control works, Centralized Version Control Systems (CVCS), Distributed Version Control Systems (DVCS), advantages of DVCS, Private Workspace, Easier merging, Easy to scale horizontally, List the disadvantages of DVCS, Explain how CVCS and DVCS compare with each other, Describe the working of the multiple repositories model Unit IV Application Containerization Understanding Containers: Transporting Goods Analogy, Problems in Shipping Industry before Containers, Shipping Industry Challenges, Container: Virtualization Introduction, Hypervisor, Scope of Virtualizations, Containers vs Virtual Machines, Understanding Containers, Containerizations Platform, Runtime and Images, Container Platform, Container Runtime, The Chroot System, FreeBSD Jails, Linux Containers (LXC), Docker

Introduction to Containerization: Docker architecture, Docker Daemon (Container Platform), Docker Rest API, CLIDifferent environments: (Dev, QA and Prod), Overcoming issues with different environments, Development Environment Docker Swarm and Kubernetes, Architecture, AWS (ECS, EKS), AWS Elastic Container Services Architecture, Azure Kubernetes Services, Openshift, Kubernetes on cloud, Monitoring of container

Laboratory Work:

Course Learning Objectives (CLO)

After the completion of the course the student will be able to:

1. Understand the benefits of DevOps over other software development processes.
2. Understand the phases of software development-delivery pipeline and automation benefits.
3. Identify and apply continuous integration and deployment prerequisites, process and benefits.
4. Understand and implement the continuous delivery engineering practices and release process.

5. Identify & use the test-driven deployment and various tools/frameworks used for continuous integration and delivery in DevOps.
6. Demonstrate the different DevOps Tools like Git, Docker, and Kubernetes etc.

Text Books

1. Arundel, J., & Domingus, J., Cloud Native DevOps with Kubernetes: building, deploying, and scaling modern applications in the cloud. O'Reilly Media, (2019).
2. Kim, G., Humble, J., Debois, P., & Willis, J., The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations. IT revolution Press 1st ed(2016).
3. Bass, L., Weber, I., & Zhu, L., DevOps: A software architect's perspective. Addison-Wesley Professional (2015).

Reference Books:

1. Fox, A., Patterson, D. and Joseph, S., Engineering Software as a Service: An Agile Approach Using Cloud Computing, 1st Edition (2013).
2. Rossel, S., Continuous Integration, Delivery, and Deployment: Reliable and Faster Software Releases with Automating Builds, Tests, and Deployment. Packt Publishing, 1st ed (2017).

Note: Color coding for NVIDIA baskets:

Teaching Kit (Audio or PPT only) / NVIDIA DLI (Online Courses)

Manual Teaching (NVIDIA/Fore Solutions)

Self-Learn From Online Material (Open Source Reading/Lectures)